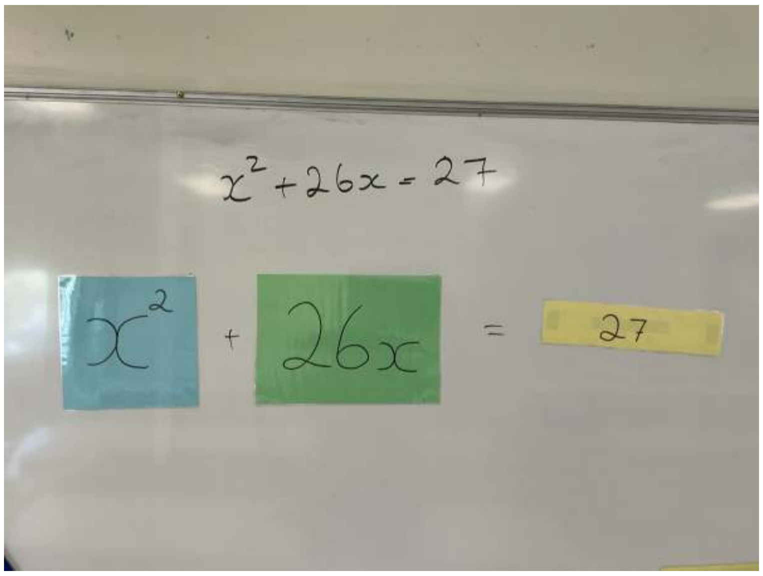
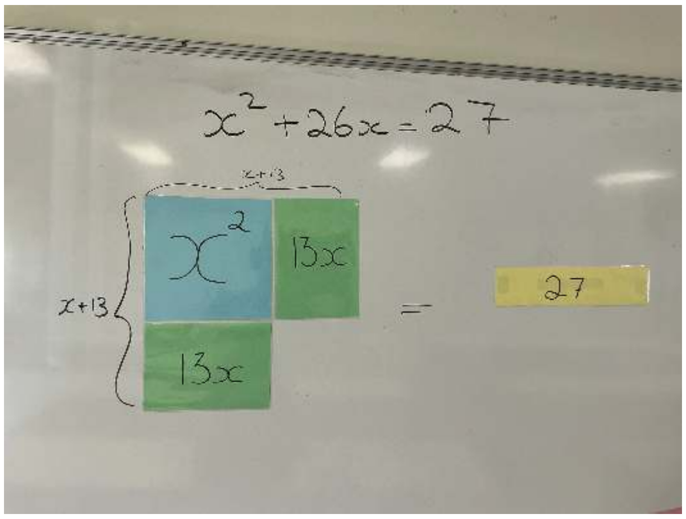
$$x^2+26x=27$$

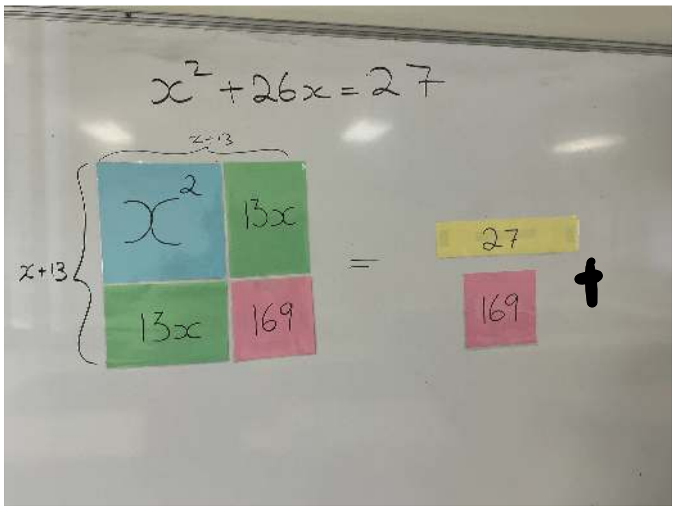
Step 1 - write the equation

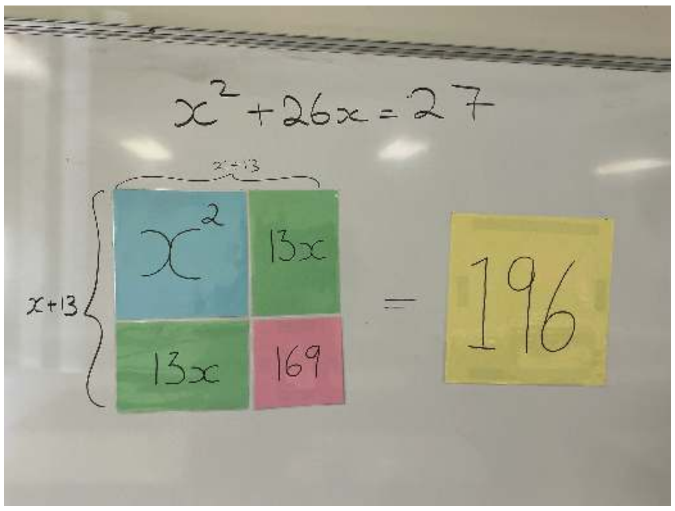
step 2 - Build, but halve $26x$





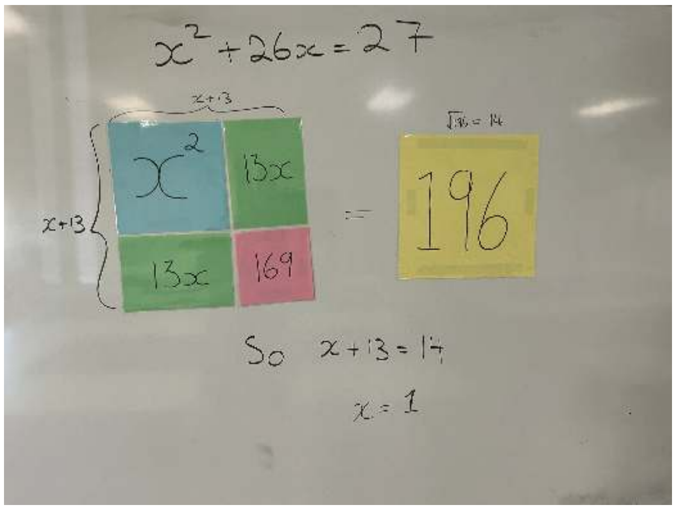
Step 3 → We "complete the square"





Step 4

Work out side lengths by taking square root.



note: During this time, the negative solution wasn't considered.

For thousands of years, mathematicians were oblivious to the negative solutions to their equations because they were working in the real world.

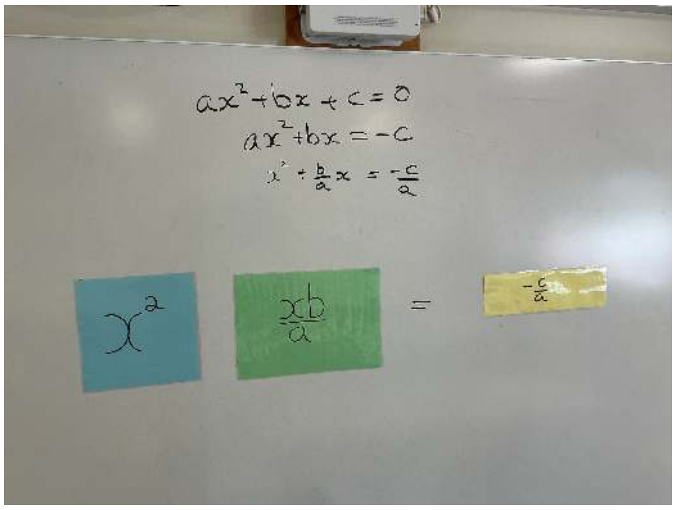
$$(x+13)^2 = 196$$

$$x + 13 = \pm 14$$

$$x = 1 \text{ or } 27$$

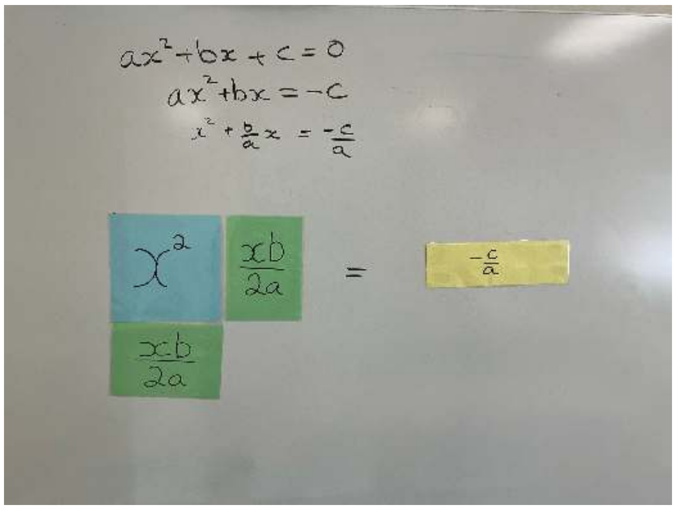
Step 5

↳ Proof of quad formula



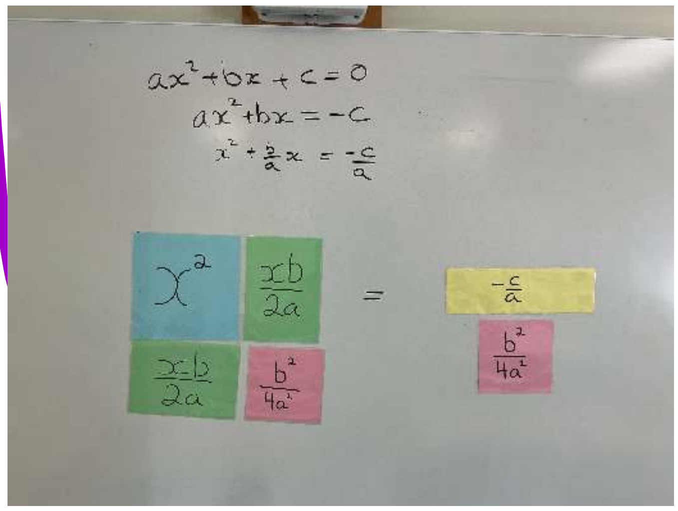
the setup.

Step 6



Step 7

complete the square



Step 8

Build → algebra to proof.